



Delhi Public School, Howrah

Pre-Board II Examination (2025 – 26)

Class-X

Care must be taken not to write anything on the question paper. All the questions must be attempted at the correct sequence.

Subject: - Mathematics Standard (Code No - 041)

Time: 3 Hours

Maximum Marks: 80

General Instructions:

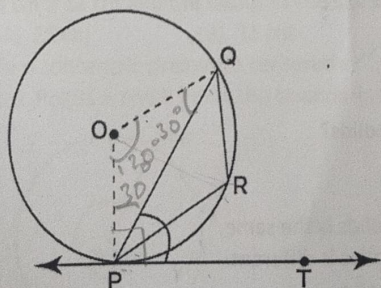
Read the following instructions carefully and follow them:

1. This question paper contains 38 questions.
2. This Question Paper is divided into 5 Sections A, B, C, D and E.
3. In Section A, Questions no. 1-18 are multiple choice questions (MCQs) and questions no. 19 and 20 are Assertion-Reason-Based questions of 1 mark each.
4. In Section B, Questions no. 21-25 are very short answer (VSA) type questions, carrying 02 marks each.
5. In Section C, Questions no. 26-31 are short answer (SA) type questions, carrying 03 marks each.
6. In Section D, Questions no. 32-35 are long answer (LA) type questions, carrying 05 marks each.
7. In Section E, Questions no. 36-38 are case study-based questions carrying 4 marks each with subparts of the values of 1, 1 and 2 marks each respectively.
8. All Questions are compulsory. However, an internal choice in 2 Question of Section B, 2 Questions of Section C and 2 Questions of Section D has been provided. An internal choice has been provided in all the 2 marks questions of Section E.
9. Draw neat and clean figures wherever required.
10. Take $\pi = 22/7$ wherever required if not stated.
11. Use of calculators is not allowed.

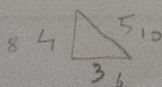
Section A

(Section A consists of 20 questions of 1 mark each.)

- 1) $(1 + \sqrt{3})^2 - (1 - \sqrt{3})^2$ is: $1+3+2\sqrt{3} - (1+3-2\sqrt{3}) = 1+3+2\sqrt{3} - 1-3+2\sqrt{3} = 4\sqrt{3}$
 (A) A positive rational number
 (B) A negative integer
 (C) A positive irrational number
 (D) A negative irrational number
- 2) The distance between two points A and B, on a graph is given as $\sqrt{10^2 + 7^2}$. The coordinates of $(-4, 3)$. Given that point B lies in the first quadrant, then all the possible x-coordinates of point B are: A
 (A) Multiple of 2
 (B) Multiple of 3
 (C) Multiple of 5
 (D) Multiple of 6
- 3) Shown below is a pair of linear equations
 $mx + 4y - 6 = 0$ and $ny - 12x + 12 = 0$
 For which of the following values of m and n do the above equations have infinitely many solutions?
 (A) $m = -1$ and $n = 2$
 (B) $m = -1$ and $n = -3$
 (C) $m = 6$ and $n = -8$
 (D) $m = 6$ and $n = -2$
- 4) In the following figure, PQ is a chord of a circle and PT is the tangent at P such that $\angle QPT = 60^\circ$. Then $\angle PRQ$ is equal to:



- (A) 135° (B) 150° (C) 120° (D) 110°
- 5) The sine of an angle in a right triangle is $\frac{4}{5}$. Which of these could be the measures of the sides of the triangles?
 (A) 4 cm, 5 cm and 9 cm
 (B) 4 cm, 5 cm and $\sqrt{41}$ cm
 (C) 6 cm, 8 cm and 10 cm
 (D) 8 cm, 10 cm and $4\sqrt{41}$ cm



- 6) If $x^2 + bx + b = 0$ has two real and distinct roots, then the value of b can be: (D) -3
 (A) 0 (B) 4 (C) 3
- 7) Mr. Sharma buys a property every year for 12 years. Every year he buys x acres more than the previous year. If in the 8th year he bought 45 acres of land and in the 5th year he bought 30 acres of land, how many acres did he buy in the last year?
 (A) 60 acres (B) 65 acres (C) 115 acres (D) 125 acres
- 8) A box contains some new and weathered cricket balls of two colours. This data is shown in the table below.

Condition \ Colour	New	Weathered
White	5	5
Red	7	3

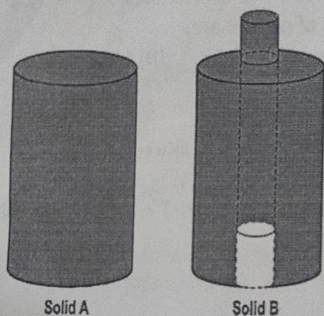
Pratik picks a new, white ball and puts it back in the box. If a ball is then picked randomly from the box, what is the probability that it is NOT the same variety as Pratik picked?

- (A) $\frac{1}{15}$ (B) $\frac{5}{20}$ (C) $\frac{5}{15}$ (D) $\frac{15}{20}$
- 9) Shown below is a solved trigonometric problem.

$$\begin{aligned}
 & \frac{\operatorname{cosec} \theta + \cot \theta - 1}{\operatorname{cosec} \theta - \cot \theta + 1} \\
 &= \frac{\operatorname{cosec} \theta + \cot \theta - (\cot^2 \theta - \operatorname{cosec}^2 \theta)}{\operatorname{cosec} \theta - \cot \theta + 1} \quad (\text{step 1}) \\
 &= \frac{\cot \theta + \operatorname{cosec} \theta - (\cot \theta - \operatorname{cosec} \theta)(\cot \theta + \operatorname{cosec} \theta)}{\operatorname{cosec} \theta - \cot \theta + 1} \quad (\text{step 2}) \\
 &= \frac{(\cot \theta + \operatorname{cosec} \theta)(1 - \cot \theta + \operatorname{cosec} \theta)}{\operatorname{cosec} \theta - \cot \theta + 1} \quad (\text{step 3}) \\
 &= \cot \theta + \operatorname{cosec} \theta \quad (\text{step 4})
 \end{aligned}$$

In which step is there an error in solving?

- (A) Step - 1 (B) Step - 2 (C) Step - 3 (D) There is no error.
- 10) If n is a natural number, then $2(5^n + 6^n)$ always ends with:
 (A) 1 (B) 4 (C) 3 (D) 2
- 11) Solid B is made using Solid A by cutting a smaller cylinder through the centre which is then pushed from below such that it protrudes from the top as shown below.

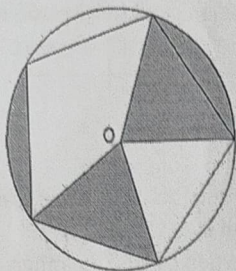


(Note: The figure is not to scale.)

Which of these is true about the volume and surface area of the two solids?

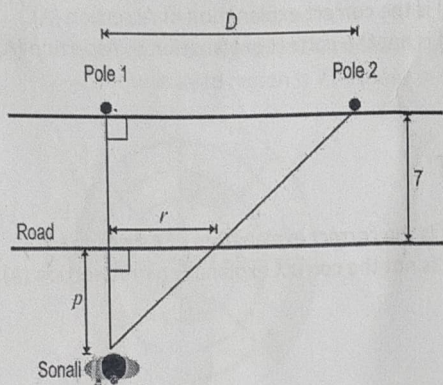
- (A) Both the volume and the surface area of the solids are different.
 (B) Both the volume and the surface area of the solids are the same.
 (C) The volume of the solids is different, but the surface area of the solids is the same.
 (D) The volume of the solids is the same but the surface area of the solids is different.
- 12) Which of the following is a quadratic polynomial having zeroes $-\frac{2}{3}$ and $\frac{2}{3}$?
 (A) $4x^2 - 9$ (B) $\frac{4}{9}(9x^2 + 4)$ (C) $x^2 + \frac{9}{4}$ (D) $5(9x^2 - 4)$
- 13) Ginny flipped a fair coin three times and tails came up each time. Ginny wants to flip the coin again. What is the probability of getting heads in the next coin flip?
 (A) 0 (B) 0.25 (C) 0.5 (D) 1

- 14) A regular pentagon is inscribed in a circle with centre O, of radius 5 cm, as shown below.



What is the area of the shaded part of the circle?

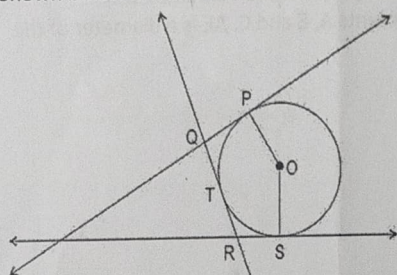
- (A) $2\pi \text{ cm}^2$ (B) $4\pi \text{ cm}^2$ (C) $5\pi \text{ cm}^2$ (D) $10\pi \text{ cm}^2$
- 15) Sonali is standing on one side of a 7 m wide road as shown below. She wants to estimate the distance(D) between two light poles on the other side without crossing the road.



(Note: The figure is not to scale. All measurements are in metres.)

Which of the following expressions represent D in terms of p and r?

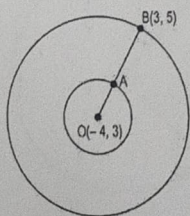
- (A) $\frac{7r}{p}$ (B) $\frac{pr}{7}$ (C) $\frac{pr}{p+7}$ (D) $\frac{r(p+7)}{p}$
- 16) Shown below is a circle with centre O having tangents at points P, T and S.



(Note: The figure is not to scale.)

If QR = 12 cm and the radius of the circle is 7 cm, what is the perimeter of the polygon PQTRSO?

- (A) 26 cm (B) 31 cm (C) 38 cm (D) Cannot say with the given information.
- 17) Two concentric circles are centered at O(-4, 3). The ratio of the area of inner circle to that of the outer circle is 1:9. Points A and B lie on the boundaries of the inner and outer circle, respectively, as shown below.



(Note: The figure is not to scale.)

The coordinates of point B are (3, 5). Which of the following are the coordinates of A?

- (A) $(\frac{2}{3}, \frac{13}{3})$ (B) $(-\frac{5}{3}, \frac{11}{3})$ (C) $(-\frac{9}{4}, \frac{14}{4})$ (D) $(-\frac{33}{10}, \frac{16}{5})$

- (18) The heights of plants in Dipti's garden are recorded in the table given below. The median plant height is 55 cm.

Heights of plants (in cm)	0-20	20-40	40-60	60-80	80-100
Number of plants	$2x$	4	$4x$	8	4

Which of the following is the value of x ?

- (A) 1
(B) 2
(C) 8
(D) the value of x cannot be found without knowing the total number of plants
- 19) A number q is prime factorized as $3^2 \times 7^2 \times b$, where b is a prime number other than 3 and 7.
Based on the above information, two statements are given below - one labelled Assertion (A) and the other labelled Reason (R). Read the statements carefully and choose the option that correctly describes statements (A) and (R).

Assertion (A): q is definitely an odd number.

Reason (R): $3^2 \times 7^2$ is an odd number.

- (A) Both Assertion (A) and Reason (R) are true, and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true, and Reason (R) is not the correct explanation of Assertion (A).
(C) Assertion (A) is true, but Reason (R) is false.
(D) Assertion (A) is false, but Reason (R) is true.

- 20) **Assertion (A):** For an acute angle θ , $\sec \theta = 3 \Rightarrow \tan \theta = 2\sqrt{2}$

Reason (R): $\sec^2 \theta = 1 + \tan^2 \theta$

- (A) Both Assertion (A) and Reason (R) are true, and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true, and Reason (R) is not the correct explanation of Assertion (A).
(C) Assertion (A) is true, but Reason (R) is false.
(D) Assertion (A) is false, but Reason (R) is true.

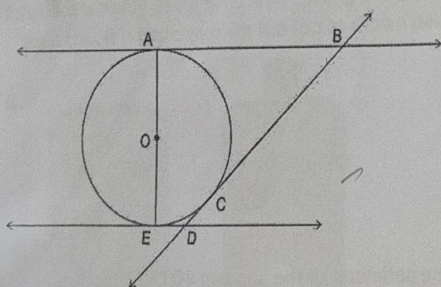
Section B

(Section B consists of 5 questions of 2 marks each.)

- 21) (A) If the m^{th} term of an AP be $\frac{1}{n}$ and n^{th} term be $\frac{1}{m}$, then show that its $(mn)^{\text{th}}$ term is 1.
OR

(B) Find the sum of all two-digit natural number which when divided by 3 yield 1 as remainder.

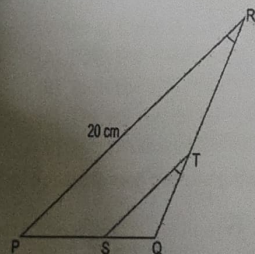
- 22) Shown below is a circle with centre O and three tangents drawn at points A, E and C. AE is a diameter of the circle. The tangents intersect at points B and D.



Based on the above information, evaluate whether the following statement is true or false. Justify your answer.

At least one pair of opposite sides of AEDB is parallel.

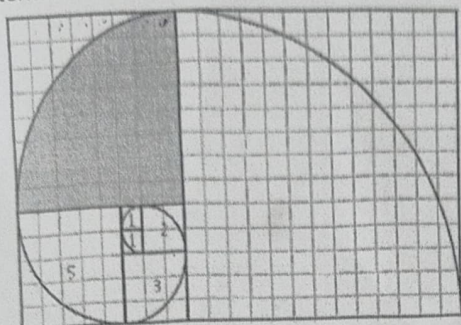
- 23) ΔPQR is shown below. ST is drawn such that $\angle PRQ = \angle STQ$.



(Note: The figure is not to scale.)

If ST divides QR in a ratio of 2:3, then find the length of ST.

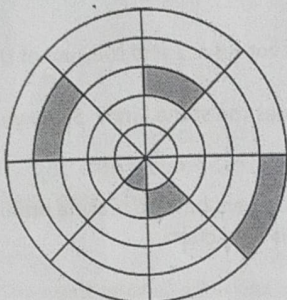
- 24) (A) Fibonacci sequence is a pattern in which each number is obtained by adding the previous two numbers (except the first 2 numbers). The pattern is 0, 1, 1, 2, 3, 5, 8, 13... Shown below is a representation of the first few terms of the Fibonacci sequence in a unit square grid. The terms represent the side lengths of the squares.



Find the area of the shaded sector.

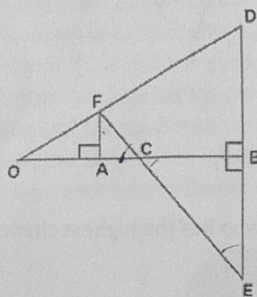
OR

- (B) Shown below is a figure such that each circle is divided into equal sectors by 4 straight lines. The combined area of the shaded region is 77 square units.



Find the radius of the largest circle? Show your work.

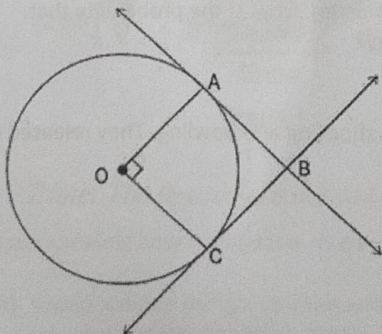
- 25) In the following figure, OB is the perpendicular bisector of DE. $AF \perp OB$. FE intersects OB at C. Prove that $\frac{OA}{OB} = \frac{AC}{BC}$.



Section C

(Section C consists of 6 questions of 3 marks each.)

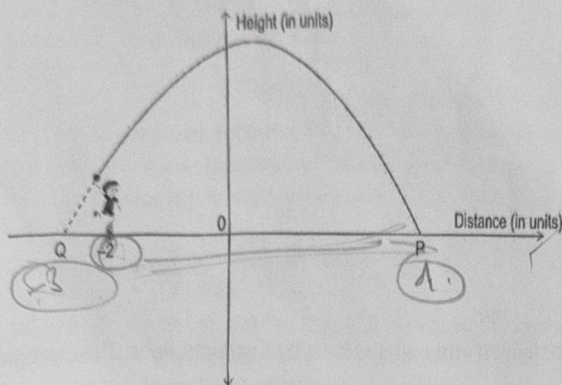
- 26) Shown below is a circle with centre O. Tangents are drawn at points A and C, such that they intersect at point B.



If $OA \perp OC$, then show that quadrilateral OABC is a square.

27) Prove that $5\sqrt{7} + 2$ is an irrational number.

28) Riddhi throws a stone in the air such that it follows a parabolic path before it lands at P on the ground as depicted by the graph below.



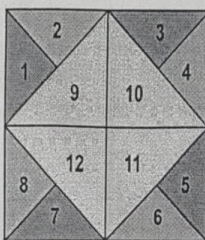
(Note: The figure is not to scale.)

- The above graph is represented by a polynomial where the sum of its zeroes is 1 and the sum of the squares of its zeroes is 25. Find the coordinates of P and Q.
- If one unit on the graph represents 25 metres, how far from Riddhi does the stone land? Show your work.

29) State and prove Basic Proportionality Theorem.

30) (A) A square dartboard has sections numbered from 1 to 12 as shown below. Players have to make a prediction and throw a dart. They win if their dart lands on the section that matches their prediction.

- Arya says, "My dart will land on a composite number."
- Bashir says, "My dart will land on an even number."
- Cathy says, "My dart will land on a factor of 12."



Calculate the probability of each of their predictions occurring and determine who has the highest chances of winning. Show your work.

OR

(B) Drish lives in India and Hugh lives in the USA. The date formats of both the countries is given below.

- India: day/month
- USA: month/day

They wrote dates every day in 2022. If a day in 2022 is randomly selected, what is the probability that:

- both their dates in the two formats are the same on that day?
- the date written by Hugh is a valid date for Drish in India?

Show your work and give valid reasons.

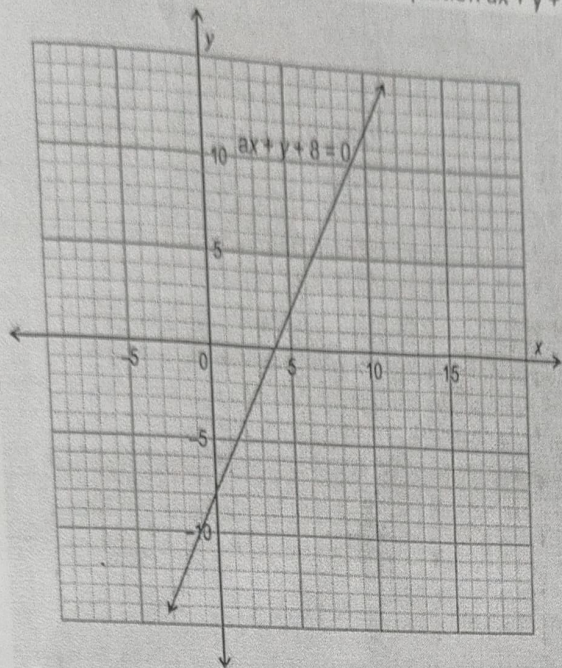
31) (A) Anuj and Safina started a new game zone consisting of two games - shooting and bowling. They released the following rate card for the customers:

Pack	Shooting	Bowling	Price (inclusive taxes)
Solo 1	-	1 round	Rs 60
Solo 2	1 round	-	Rs 75
Combo 1	3 rounds	2 rounds	Rs 285
Combo 2	4 rounds	5 rounds	Rs 485

The price of shooting is the same in both the combos, and the price of bowling is the same in both the combos. How much more is the price for one round of bowling in the solo pack than in the combo packs? Show your work.

OR

- (B) The graph of a line represented by the equation $ax + y + 8 = 0$ is shown in the figure below.



- Find the value of a .
- Find the point of intersection of this line with the line represented by the equation $4x - 3y - 14 = 0$. Show your work.

Section D

(Section D consists of 4 questions of 5 marks each.)

- 32) (A) The numerator of a fraction is 3 less than its denominator. If 2 is added to both numerator and denominator, then the sum of the new fraction and original fraction is $1\frac{9}{20}$. Find the original fraction.

OR

- (B) A plane left 30 minutes late than its scheduled time and to reach the destination 1500 km away in time, it had to increase its speed by 100 km/h from the usual speed. Find its usual speed.

- 33) Prove that: $(1 + \tan x + \sec x)^2 = 2(1 + \sec x)(\sec x + \tan x)$

- 34) (A) A restaurant stores ice-cream in a box with a dispenser attached for filling ice-cream cones. The dimensions of the box and the ice-cream cones used by the restaurant are shown in Figure 1 below. To make each serving of dessert, the cone is first filled with ice-cream and then topped with a hemispherical scoop of ice-cream taken from the same box, as shown in Figure 2.

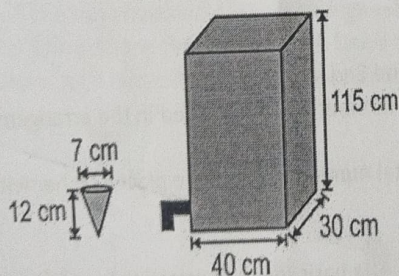


Figure 1



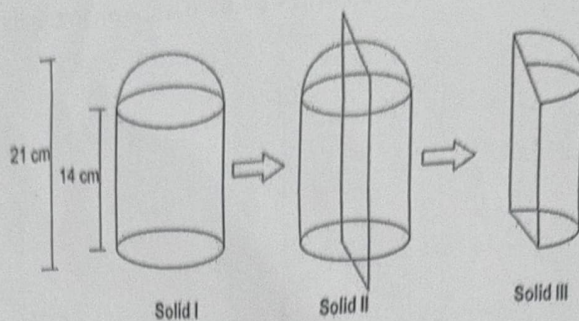
Figure 2

(Note: The figures are not to scale.)

Approximately how many desserts can be served out of a completely filled box of ice-cream? Show your steps.

OR

- (B) A solid is made using a cylinder and a hemisphere. It is then dipped in red paint and cut vertically along the axis of symmetry as shown below. The unpainted surface of solid III is then painted yellow.



(Note: The figures are not to scale.)

Find the surface area of solid III that is painted: (i) red and (ii) yellow. Show your work.

35) Observe the given table and answer the following questions:

Classes	4 - 8	8 - 12	12 - 16	16 - 20	20 - 24	24 - 28	28 - 32	32 - 36
Frequency	2	12	15	25	18	12	13	3

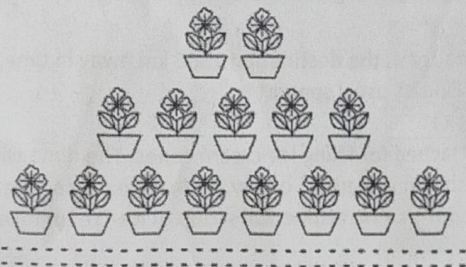
- Find the Mean of the above data by step-deviation method. *19.92*
- Find the Mode of the above data.
- Find the Median of the above data using Empirical Relation. *2 Median =*

Section E

(Section E consists of 3 case study-based questions of 4 marks each.)

36) **Case Study - 1**

Aahana being a plant lover, decides to convert her balcony into beautiful garden full of plants. She bought few plants with pots for her balcony. She placed the pots in such a way that number of pots in the first row is 2, second row is 5, third row is 8 and so on.



Based on the above information, answer the following questions:

- Find the number of pots placed in the 10th row.
- Find the difference in the number of pots placed in 5th row and 2nd row.
- If Aahana wants to place 100 pots in total, then find the total number of rows formed in the arrangement.

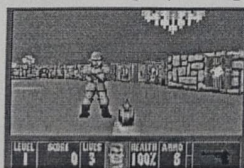
OR

- If Aahana has sufficient space for 12 rows, then how many total numbers of pots are placed by her with the same arrangement?

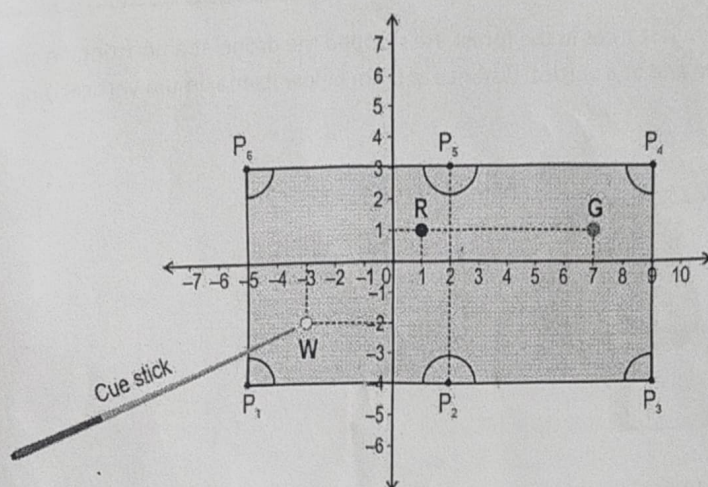
37) **Case Study - 2**

Raycasting is a technique used in the creation of computer games. The basic idea of raycasting is as follows: the map is a 2D square grid. Using rays generated from an object, this 2D map can be transformed into a 3D perspective. One of the methods involves sending out a ray from the player's location. To determine how far he/she is from a wall or an obstacle, the distance between the player's coordinates and the coordinate of the wall is calculated. If the player is near the obstacle, it looks larger and vice-versa.

Shown below is a game, Wolf 3D, which was created using raycasting.



Riju wants to create an online snooker game using raycasting. The game in the creation stage on a coordinate map is shown below.



- How much distance will a ray travel if sent from the green ball to the nearest pocket? Show the steps.
- Riju wants to place a yellow ball at the midpoint of the line connecting the white and green balls. Find the coordinates of the point at which he should place the yellow ball. Show the steps.
- Riju is running a trial on his game. He struck the white ball in a way that it rebounded off the rail (line connecting P_4 and P_6) and went into the pocket P_2 .
 - After the rebound, the ball crossed the x-axis at point $X\left(\frac{2}{7}, 0\right)$ on the way to the pocket.
 - The ratio of the distance between the rail and point X and the distance between point X and the pocket was 3 : 4.

Find the coordinates of the point at which the ball struck the rail. Show your steps.

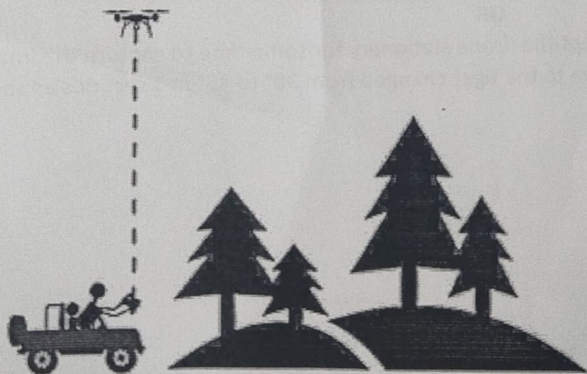
OR

- Riju wants to hit a blue ball placed at $(-1, -3)$ into pocket P_5 along a straight path. Would the red ball lie on the straight path between the blue ball and P_5 ? Justify your answer.

38) Case Study – 3

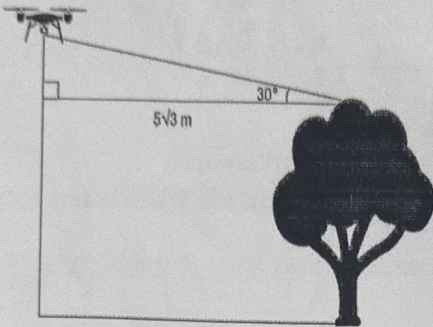
A drone is an aircraft without any human pilot and is controlled by a remote-controlled device. Its various applications include policing, surveillance, photography, precision agriculture, forest fire monitoring, river monitoring and so on.

David used an advanced drone with high resolution camera during an expedition in a forest region which could fly up to 100 m above the ground level. David rode on an open jeep to go deeper into the forest. The initial position of drone with respect to the open jeep on which David was riding is shown below.



David's jeep started moving to enter the forest at an average speed of 10 m/s. He Simultaneously started flying the drone in the same direction as that of the jeep.

- (a) David reached near one of the tallest trees in the forest. He stopped the drone at a horizontal distance of $5\sqrt{3}$ m from the top of the tree and at a vertical distance of 65 m below its maximum vertical range.



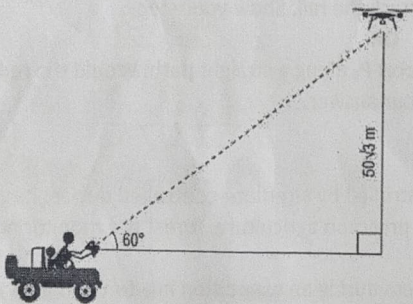
(Note: The figure is not to scale.)

If the angle of elevation of the drone from the top of the tree was 30° , find the height of the tree. Show your work.

- (b) The drone was flying at a height of $30\sqrt{3}$ metres at a constant speed in the horizontal direction when it spotted a zebra near a pond, right below the drone. The drone travelled for 30 metres from there and it could see the zebra, at the same place, at an angle of depression of θ from it.

Draw a diagram to represent this situation and find θ . Show your work.

- (c) After 2 minutes of starting the expedition both the drone and the jeep stopped at the same moment so that the drone could capture some images. The position of the drone and the jeep when they stopped is as shown below.

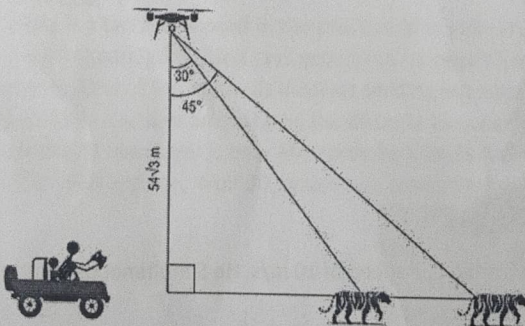


(Note: The figure is not to scale.)

Find the average speed of the drone in m/s rounded off up to 2 decimal places. Show your work.

OR

- (d) At some point during the expedition, David kept the drone stationary for some time to capture the images of a tiger. The angle of depression from the drone to the tiger changed from 30° to 45° in 3 seconds as shown below.



(Note: The figure is not to scale.)

What was the average speed of the tiger during that time? Show your work. (Note: Take $\sqrt{3} = 1.73$)